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POTASSIUM ION ACCUMULATION NEAR A PACE-MAKING CELL OF *APLYSIA*

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SUMMARY

1. A delayed current decrease associated with prolonged depolarization was studied in R_{15} (the parabolic burster) of *Aplysia* by using intracellular recording and voltage clamp techniques.

2. For long duration command pulses (3 sec), the outward current shows a delayed decrease. The current goes from a maximum near 100 msec and falls until a steady-state outward current is reached between 1·5 and 2·5 sec after the beginning of the command step. This final steady-state current is usually only about 20–30% of the peak outward current.

3. Double step voltage clamps show that this current decrease is associated with a large shift of e.m.f. Measurements of conductance, on the other hand, fail to show any significant difference in conductance associated with peak and steady-state currents.

4. From the results of application of high K^+ ringer, the conclusion is reached that this shift in e.m.f. is due to an accumulation of K^+ near the exterior cell membrane. Several other experiments exclude the possibility of either metabolic events or compensating conductance changes producing the phenomenon.

5. The location of the accumulation is considered on the basis of anatomical studies. It is concluded that the accumulation takes place in the extensive infoldings found in cells like R_{15}. An explanation of the difference in delayed current decrease between pace-makers and non-pace-makers is suggested, since the pace-makers apparently have more extensive invaginations than the non-pace-makers. This suggestion is lent support by measurements of capitance and current density.

INTRODUCTION

Since its first practical application, voltage clamp technique has proved to be an exceptionally useful tool for probing the mechanisms of excitable tissue. At first, for voltage commands of relatively short duration, it appeared that the membrane properties of nerve and muscle cells could be described by a system of relatively simple parameters (Hodgkin & Huxley, 1952*a*, *b*, *c*). It seemed that most systems could be represented by three basic phenomena; that is, first a conductance increase to sodium ion (Na^+ activation) followed by a conductance increase to K^+ (K^+ activation) and a decline in the large Na^+ conductance (Na^+ inactivation). However, as voltage clamp studies were extended by using longer or more complex voltage command steps, it became clear that the three simple parameters described for short pulses were inadequate to explain the observed changes in current. Probably the first phenomena of this type was described in the pulmonate mollusc, *Onchidium* (Hagiwara & Saito, 1959; Hagiwara, Kusano & Saito, 1961). In this preparation, not only were the normal early inward and outward currents described corresponding to the Hodgkin & Huxley parameters, but also a second K^+ conductance pathway was found whose activation properties were different from the normal outward current pathway. Moreover, in several cases the normal K^+ outward current displayed a delayed current decrease that appeared to be K^+ inactivation. Subsequent to this first report, several other preparations were investigated and found to show delayed current decreases or other current changes not readily explained by the simple parameters mentioned above. In supramedullary cells of puffer, a delayed outward current decrease was found (Nakajima & Kusano, 1966). This was attributed to an inactivation of the K^+ outward current channel. Other changes in current were attributed to the activation and subsequent inactivation of a second K^+ pathway. The correlation between this second pathway and that observed in *Onchidium* was not clear. At approximately the same time K^+ inactivation was described in squid giant axon (Ehrenstein & Gilbert, 1966). Here the inactivation was a conductance decrease to K^+ associated with a delayed current decrease similar to that observed in puffer. It was shown that the Hodgkin–Huxley equations could predict the current changes by inclusion of a fourth parameter. More recently, both in the nudibranch, *Anisodoris* (Connor & Stevens, 1971) and in frog skeletal muscle (Adrian, Chandler & Hodgkin, 1970*a*, *b*) similar delayed current decreases have been reported. As in the case of puffer and squid, these changes were explained in terms of K^+ conductance changes taking place at times that were long compared to the early inward currents.

In contrast to these findings that most late current changes were produced by conductance changes, the delayed current decrease in *Aplysia* cells displays several differences that are not readily explained on the basis of a conductance change to a single ion like K^+. Pace-making cells of the preparation display a marked current decrease associated with a large reversal potential change while the non-pace-making cells show no such decrease. There is no apparent change in conductance that could account for the delayed current decrease in the pace-makers (Alving, 1969). The object of this paper is to determine the mechanism of this delayed current decrease and contrast it with those found in other preparations as well as suggesting differences between the pace-making and non-pace-making cells that could account for the differences in their current–voltage relations.

METHODS

Preparation. The experimental animal was the tectibranch mollusc. *Aplysia californica* (Cooper). Specimens were collected along the Pacific Coast between San Diego and Los Angeles. The specimens used varied in weight between 100 and 750 g. The abdominal ganglion of the animal was removed and placed in a Lucite chamber where it was secured by placing a platinum ring firmly over the connectives. The preparation was immersed in a modified artificial sea water (Kikuchi, Naito, & Tanaka, 1962) whose composition is given in Table 1. Table 1 also lists the composition of the other experimental salines used. A pH of 7·8 was maintained in all of these solutions with Tris buffer. The ganglion was illuminated from below through a fibre optics system. The cell R_{15} (Frazier *et al.* 1967) sometimes referred to as the parabolic burster (Strumwasser, 1965) was chosen as a suitable example of a pace-maker cell and was used throughout the course of these experiments. The connective tissue over the cell was removed with fine scissors exposing the cell to the external saline.

TABLE 1. Salt concentrations (mM)

Ringer	NaCl or $Na\ MeSO_3$	KCl or $K\ MeSO_3$	$CaCl_2$ or $Ca(MeSO_3)_2$	$MgCl_2$ or $Mg(MeSO_3)_2$	Tris buffer	TEA
Normal	420	10	10	25	100	—
Na^+-free	0	10	10	25	520	—
Cl^--free*	420	10	10	25	100	—
High K^+	330	100	10	25	100	—
TEA	370	10	10	25	100	50

* Cl^--free solutions contain methanesulphonate as the anion.

Electrodes. Glass micro-electrodes filled with 5 M K acetate were used for recording and current injection. K acetate was used rather than KCl for two reasons: first, there has been some suggestion that intracellularly injected chloride ion may cause changes in membrane properties (Strumwasser, 1965). The use of acetate as the anion obviates the question of intracellular chloride effects of this sort. In addition, the electrical properties of these electrodes seem slightly more suitable than KCl

electrodes. The resistance of K acetate electrodes is slightly less for comparable tip diameters; moreover, the current passing ability is generally more linear allowing larger currents for voltage control. The voltage and reference electrodes were selected and matched with resistances of 5–7 MΩ. The current electrode was 1–2·5 MΩ and was usually selected to pass 10–50 μA in the extracellular saline for 2 sec. To reduce capacitive coupling between current and voltage electrodes two alternative methods were used: either a grounded foil shield was inserted between the two electrodes or the current electrode was coated with silver bearing epoxy circuit paint and the coating grounded.

Electrical recording. The circuit used for voltage clamping was essentially the same as those described by Hagiwara, Takahashi & Junge (1968) and Brown *et al.* (1970). The voltage feed-back amplifier was a Tektronix type D differential plug-in unit. The membrane potential was fed through ELSA unity gain amplifiers while the membrane current was monitored by means of a virtual ground system utilizing an operational amplifier. With the system tuned, the rise of the membrane potential for a square command pulse was 200–250 μsec. Between command pulses the potential was maintained at the most negative potential observed during the normal activity cycle. Subsequently this potential will be referred to as the resting potential. At periodic intervals the voltage clamp was turned off. This allowed a check of the stability of the holding potential as well as an opportunity to monitor the condition of the cell and its ability to generate a normal activity cycle.

Two and occasionally three microelectrodes were inserted into the cell. Penetration was considered satisfactory when all of the electrodes recorded the same potential and the cell displayed normal burst activity (Strumwasser, 1965). In the cases where three electrodes were inserted, the third electrode was used to determine the isopotentiality of the cell. In the cells checked, all but the largest cells (greater than 400 μ) or cells driven at extreme potentials (greater than 120 mV above resting potential) displayed good isopotentiality (within 5 %) at all points checked in the cell soma. To eliminate problems due to inadequate spatial potential control, an effort was made to use only relatively small cells and command potentials not greater than 120 mV above resting potential. It should be remarked here that the potential of the axon was probably not constant for any great distance from the soma. Evidence of this is suggested by the small inward current inflexions in the current time records at high gain. These were largely disregarded as small compared to the soma currents under consideration.

RESULTS

Fig. 1 shows the normal current–time and current–voltage characteristics of R_{15} for command pulses of approximately 3 sec duration. The current–voltage curves in Fig. 1 are taken at 140 msec after the beginning of the command step which corresponds to the time of peak outward current (I_{peak}, abbreviated I_p) and at 2800 msec after the current has decreased to a final steady value ($I_{steady\ state}$, abbreviated I_{ss}). At the time scale of the current–time curves the early inward current is not apparent. The outward current reaches a maximum between 100 and 140 msec after initiation of the command potential. It then begins to decrease rapidly until it reaches a plateau at about 1500 msec which may be only 20–30 % of the maximum outward current. The phenomenon is first

apparent at a command potential of approximately -20 mV. If the ratio of peak current to steady-state current is used as a measure of the phenomenon, then the amount of current decay reaches a saturation level at command potentials of approximately $+10$ mV.

To study the conductances more quantitatively and also to gain information about the reversal potentials for the currents involved, a double command step voltage clamp was undertaken. This technique is based on consideration of the expression

$$I_T = g\,(V - E_R)$$

where I_T is the total membrane current, g is the apparent membrane conductance, V is the membrane potential, and E_R is the reversal potential

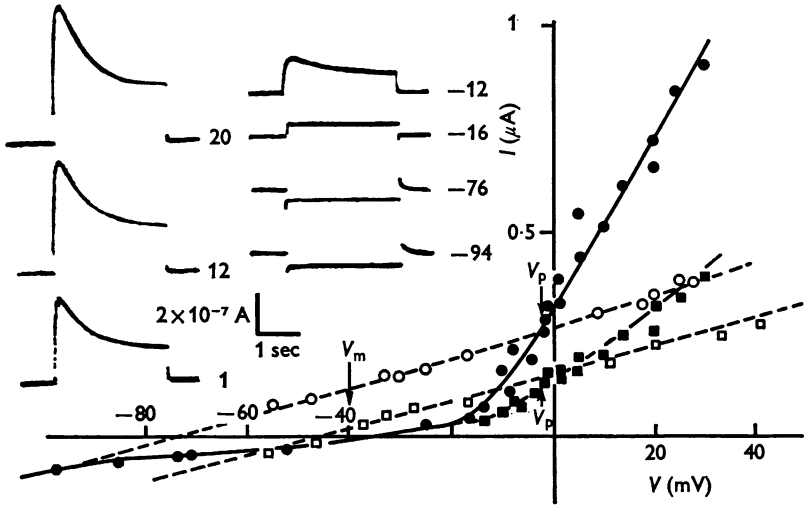


Fig. 1. Normal and instantaneous I–V relationships. The normal I–V relationship at the time of peak outward currents (filled circles) and steady-state currents (filled squares) is shown. The instantaneous I–V relationship at a primary command voltage (V_p) for peak (open circles) and steady-state currents (open squares) are superimposed. Current time records at various potentials are given in the inset.

for the membrane currents. The expression represents an equation for a straight line if g is constant.

In *Aplysia*, if this property of g is true for any given voltage, then the instantaneous I–V relationship should be linear for square secondary voltage steps from the primary command potential. Moreover, the slope of the instantaneous I–V relationship would then be the equivalent conductance of the cell at the primary command voltage, while the intercept should be the reversal potential of the currents flowing at the time of the

secondary step. The results of such an experiment are superimposed on the normal I–V relationship shown in Fig. 1. The primary command was −2 mV. The secondary command pulse (100 msec) was applied either at the time of maximum outward current or at a time after attaining steady-state current. The linearity of the instantaneous I–V relationship is satisfactory with correlation coefficients greater than 0·95. With the linearity verified, the method may be used to determine the membrane conductances and reversal potentials. As mentioned above, the slope of the instantaneous relationship should be the conductance at the primary command potential, while the intercept is the reversal potential of the currents flowing at the time of the secondary step. The interesting point to notice is the shift in reversal potential associated with the change from peak to steady-state current. The reversal potential is about −70 to −90 mV at the time of peak outward current, shifting 30–40 mV more positive as the current decays to the final steady-state level at the end of the command pulse.

TABLE 2. Reversal potentials and conductance ratios at peak current and during the steady-state current

Cell no.	Saline	e.m.f. (mV) (peak)	e.m.f. (mV) (final)	g × 10⁶mho (peak)	g × 10⁶mho (final)	$\frac{g_{peak}}{g_{final}}$
P-73	Normal	−72	−41	4·1	4·0	1·0
P-116	Normal	−74	−58	3·6	3·1	1·2
P-112	Normal	−72	−23	3·1	3·8	·8
	Na^+-free	−76	−25	4·1	3·6	·9
P-138	Normal	−92	−23	4·1	3·1	1·3
	Na^+-free	−94	−25	3·6	2·8	1·3
P-131	Normal	−79	−45	3·2	2·7	1·2
	Cl^--free	−84	−46	3·1	2·6	1·2
P-112	Normal	−72	−23	3·1	3·8	·8
	High K^+	−14	+7	2·0	2·0	1·0
P-139	Normal	−99	−22	6·5	5·1	1·3
	High K^+	−42	+7	7·4	7·4	1·0
P-152	Normal	−76	−17	1·6	1·2	1·3
	High K^+	−35	−21	6·5	6·5	1·0
P-132	Normal	−94	−54	1·9	1·8	1·1
	TEA (50 mM)	−40	−40	1·0	0·98	1·0

On the other hand, the conductance as estimated from the slope at the time of the peak current is not significantly different from the conductance at the time of the steady-state current (numerical values for E_R and g are given in Table 2).

The presence of the reversal potential shift can also be demonstrated in an experiment without the use of a double step clamp. If the membrane is allowed to return rapidly to an unclamped state, the membrane potential immediately after cessation of the voltage clamp represents the membrane potential at zero membrane current which should agree with the reversal potential for membrane currents at the end of the voltage clamp. By doing this experiment at various times after the beginning of a large depolarizing command step, the shift in reversal potential may be qualitatively monitored. The results of such an experiment are depicted in Fig. 2, where the membrane reversal potential is seen to shift from a value of -70 to -30 mV. Three situations have been suggested (Alving, 1969) that could produce the observed effects.

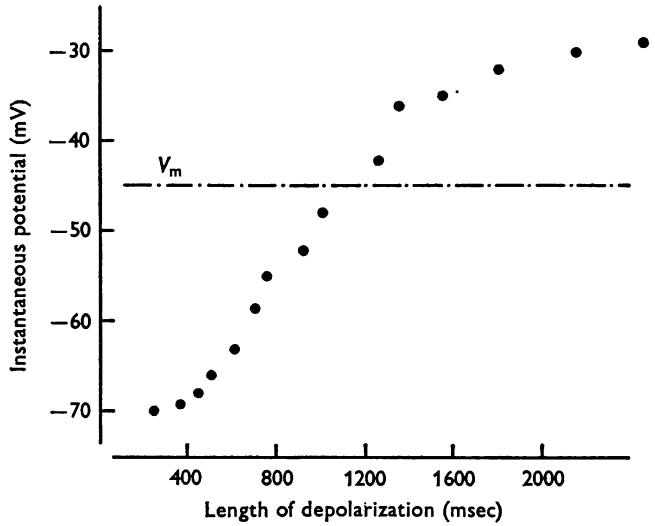


Fig. 2. Shift of post-clamp potential after depolarizing command potentials of varying length. The ordinate represents the potential immediately after cessation of membrane potential control. The abscissa is the duration of the depolarizing command step to +10 mV before the cessation of membrane voltage control. The interrupted line represents the normal membrane potential described in the text.

A. Accumulation of an ionic species. There may be an accumulation of some ion species such as K^+ near the surface of the membrane. This accumulation would cause a reduction in driving force for outward current with a consequent decrease in current even though there is no change in conductance. Accumulation in a restricted extracellular space has been described previously for squid axon (Frankenhaeuser & Hodgkin, 1956). In this case a permeability barrier is formed by the satellite cells about

150–200 μ away from the membrane surface (Geren & Schmitt, 1954). However morphological examination of the parabolic burster (Rosenbluth, 1963; Coggeshall, 1967) does not lend support to the hypothesis that there is a similar space near the *Aplysia* neurones. The satellite cells when present do not form a consistent barrier around the cells but are rather randomly interspersed among the neurones. Even more difficult to reconcile with this hypothesis is the apparent difference in current records under voltage-clamp conditions between cell types. Cells which have as many accessory cells near them as the burster do not display similar time dependent current decay (Alving, 1969).

However, other forms of diffusion limiting effects may be important in this cell. The cell surface is extensively invaginated (Bullock, 1961; Rosenbluth, 1963; Coggeshall, 1967). If exchange with the external solution is slow, extensive accumulation may occur at high current densities.

B. Compensating conductance changes. This case would require one conductance pathway whose reversal potential was relatively negative (such as K^+ in squid or Cl^- in frog muscle) and another conductance pathway whose reversal potential was relatively positive (such as Na^+ in most preparations studied). The conductance to the first pathway would predominate at the time of peak outward current. This conductance would subsequently decrease while the second conductance would increase. Since the apparent reversal potential of the membrane is the weighted sum of the e.m.f.s from the various ionic conductances, the membrane e.m.f. would become more positive with a consequent decrease in current as the second conductance pathway contributed more to the total membrane conductance. This situation would produce a large change in equivalent membrane reversal potential but the total apparent membrane conductance might change very little. In cardiac Purkinje fibres during the interspike interval, the relative Na^+ conductance is increasing while the K^+ conductance is decreasing (Trautwein & Kassebaum, 1961). The changes are not of exactly equal magnitudes so changes in equivalent membrane conductance may still be detected. If a similar situation were present in *Aplysia*, with conductance changes of comparable magnitude, the observed I–V relationships might be accounted for.

If major changes in the e.m.f.s of Na^+ or Cl^- could be made, the validity of the hypothesis could be ascertained. For example, if the Na^+ reversal potential were made large and negative (below E_K), one would expect that a suddenly increased contribution by this ion to the membrane conductance would produce a negative shift of the total membrane reversal potential.

C. Metabolic activity in the membrane producing electrogenic effects. The last possibility involves active metabolic processes which display potential

dependence. Membrane metabolic processes such as active Na^+ extrusion may be the source of significant current in *Aplysia* cells (Carpenter & Alving, 1968).

For the case of active metabolic Na^+ extrusion, the *apparent* current for a functioning system would *appear* to be outward under voltage clamp conditions. To produce the observed effects of current decrease, this metabolic activity would have to decrease when the cell was depolarized. In one case (Nakajima & Takahashi, 1966) of an apparently voltage dependent metabolic Na^+ extrusion, the time course of the current changes and the direction of the voltage dependence make a similar process in *Aplysia* a possible mechanism for producing the delayed current decrease. There has also been a suggestion that there may be active Cl^- transport in this cell that accounts for part of the activity cycle (Strumwasser, 1965).

The effect of such an active metabolic process on membrane conductance and e.m.f. is not immediately obvious. Conductance changes have been theoretically predicted (Rapoport, 1970), but there is no experimental evidence to support the theory that there must be an apparent conductance change or that there might be changes in e.m.f. associated with an active electrogenic transport process. Therefore the possibility of an active metabolic process producing the current decay and the e.m.f. shift without an apparent conductance change cannot be excluded *a priori*.

A. Na^+-free and Cl^--free salines. The second possibility mentioned above may be tested by varying E_{Na} and E_{Cl}. Appropriate changes in E_{Na} and E_{Cl} may be produced by application of salines free of these ions. In the case of Na^+, application of Na^+-free Ringer causes the Na^+ reversal potential to become large and negative (as ascertained from instantaneous voltage clamp measurements). This should result in a negative shift in membrane e.m.f. for long duration depolarizing command pulses if the Na^+ conductance pathway begins to contribute significantly at late times to the total membrane conductance. Also it follows that if a positive shift in e.m.f. in the normal case produces a decrease in outward current, a negative shift will produce a delayed current increase.

An I–V relationship produced after perfusing the preparation with Tris substituted Na^+-free saline for 1 hr is shown in Fig. 3. Qualitatively, the I–V relationship in Na^+-free saline is not significantly different from that found in normal saline. There is no change in the delayed current decrease and the positive shift in membrane reversal potential determined from instantaneous currents as one goes from the time of peak to steady-state currents is essentially unchanged. Consequently the possibility of a Na^+ conductance increase producing the delayed current decrease may be excluded.

In a similar experiment, the possibility of a Cl^- conductance change may be tested. The effect on the $I-V$ relationship of perfusing the cell for 1 hr with a saline in which all the Cl^- has been replaced with methanesulfonate ion is illustrated in Fig. 4. Cl^--free Ringer produces a slight decrease in membrane conductance. This decrease in conductance causes a

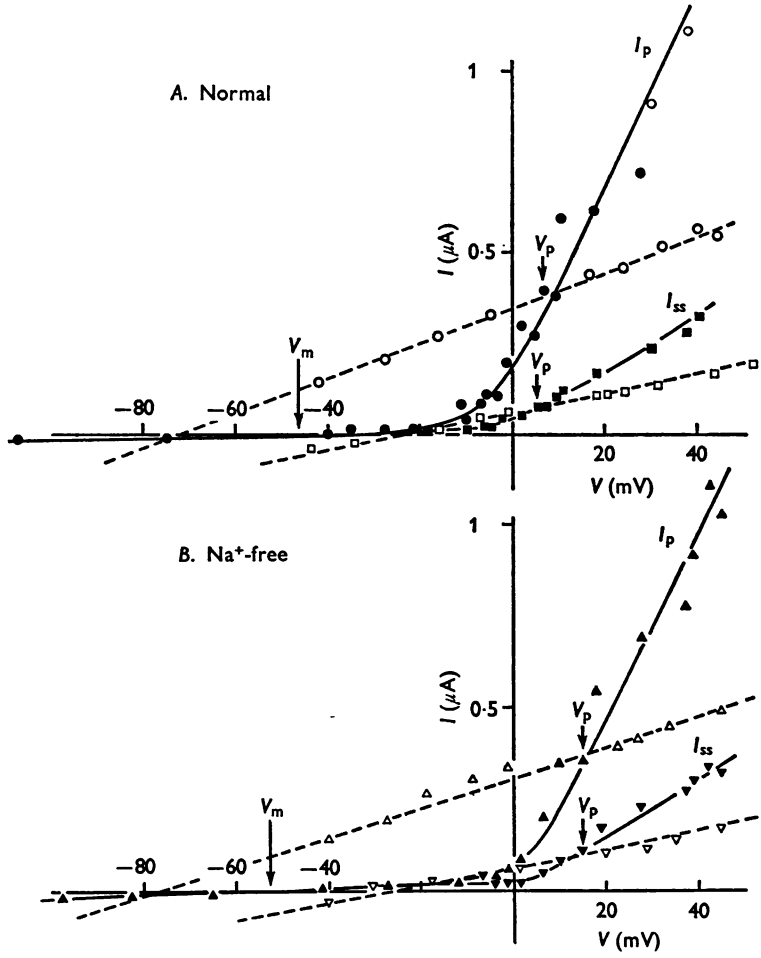


Fig. 3. Effect of Na^+-free saline. The normal and instantaneous $I-V$ relationships for peak current (filled and open circles respectively) and steady-state currents (filled and open squares) are represented in A for an experiment similar to that shown in Fig. 1. In B, the $I-V$ relationships of the same cell in Na^+-free saline is shown. Peak normal and instantaneous currents are represented by filled and open triangles respectively, while steady-state normal and instantaneous currents are given by filled and open inverted triangles.

reduction in both peak and steady-state currents. This is not an unexpected result for a membrane that displays a rather high permeability to Cl^-. The difference between the normal $I-V$ curve and the $I-V$ relationship in Cl^--free Ringer is a measure of the contribution by Cl^- to the delayed current decrease in the cell. In Cl^--free Ringer the difference between peak and steady-state currents is slightly reduced from the decrease in normal ringer. This implies some small contribution by Cl^-

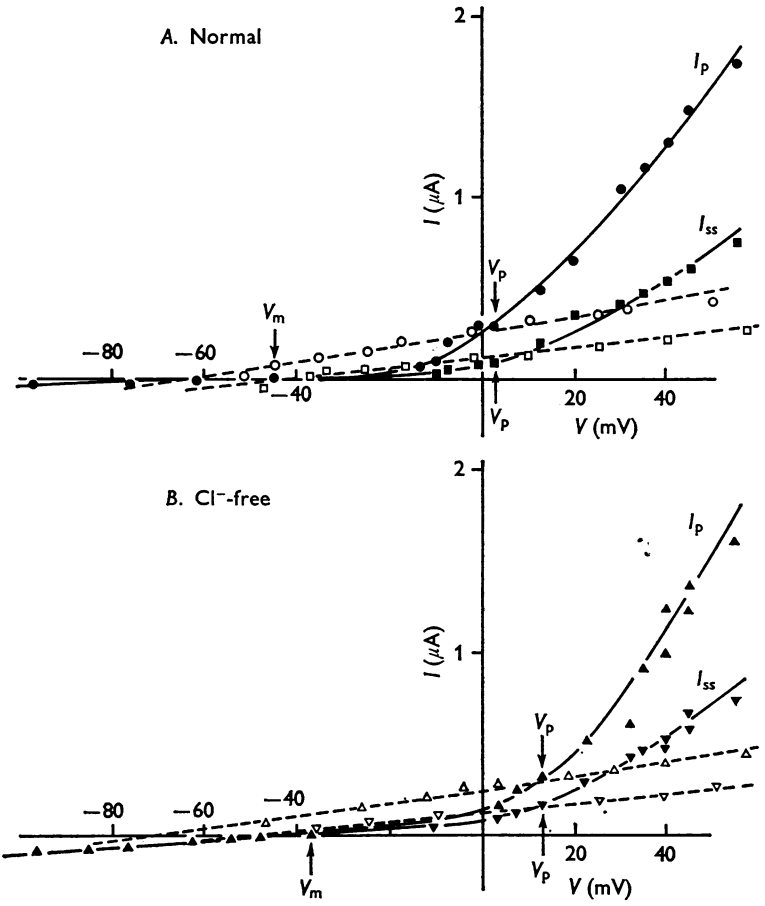


Fig. 4. Effect of Cl^--free saline. The normal and instantaneous $I-V$ relationship for peak current (filled and open circles respectively) and steady-state currents (filled and open squares) are represented in *A* for an experiment similar to that shown in Fig. 1. In *B*, the $I-V$ relationships of the same cell in Cl^--free saline is shown. Peak normal and instantaneous currents are represented by filled and open triangles respectively, while steady-state normal and instantaneous currents are given by filled and open inverted triangles.

to the delayed current decrease. However, since in Cl^- free the ratio of peak to steady-state conductance is the same as normal Ringer, this contribution cannot involve a compensating conductance change of K^+ and Cl^-.

The results of these experiments effectively rule out the second possible explanation mentioned above; that is, that the delayed current decrease is produced by an increase in conductance to Na^+, with an associated decrease in conductance to K^+ or Cl^-.

B. Metabolic inhibitors. The third possibility suggested above was that the observed slow decay of outward current might be due to a voltage dependent metabolic process producing electrogenic effects. Possible contributions of metabolic processes were determined in several ways. Active metabolic Na^+ extrusion was blocked by 10^{-4} M ouabain and suppressed by application of K^+ free or Li^+ Ringers. In all three cases, there was no change in the current–voltage relationship. These results preclude any significant contribution of active metabolic extrusion of Na^+ to the production of the delayed current decrease.

Metabolically active Cl^- processes may be eliminated after consideration of the previous experiment in Cl^--free saline. Since the cell displays a relatively high permeability to Cl^-, the internal chloride may be washed out by continuous exposure to Cl^--free Ringer. In the Cl^--free experiment the cell was perfused with Cl^--free Ringer for 1 hr before recording data. After this time the cell had reached a constant conductance implying that the Cl^- washout was complete. With both internal and external Cl^- removed, active metabolic Cl^- processes should be essentially zero. As mentioned above, the change in the I–V relationship in Cl^--free Ringer implies no contribution of Cl^- conductance to the production of the delayed current decrease. Similarly, this result also implies no contribution of active metabolic Cl^- processes contributing to the decrease.

Other metabolic processes were suppressed by reduction of the temperature of the bath to 8° C: the peak membrane outward current displayed a Q_{10} of 1·8 which is comparable to the simple conductance changes produced by reduced temperature in other similar preparations (Gorman & Marmor, 1970*a*, *b*). There is no evidence of the rather large changes associated with the effects of temperature reduction on active metabolic processes in membranes (Gorman & Marmor, 1970*b*; Carpenter & Alving, 1968). In fact, the current decay during the first 1500 msec after the potential step is nearly exponential (regression coefficients greater than 0·93). If the time constants for this decay are compared, then the decrease has a Q_{10} of 1·2 between 14 and 25° C. Values for Q_{10}'s of this size are generally associated with diffusion limited processes rather than active metabolic systems (Frankenhaeuser & Hodgkin, 1956).

C. High potassium and tetraethyl ammonium. In testing the remaining hypothesis, that there is an accumulation of ions near the membrane, one need only consider the possibility of K^+ accumulation. Consideration of the concentration gradients for Na^+ and Cl^- as well as the results of the Na^+-free and Cl^--free experiments preclude any major contribution of these ions. For the case of K^+ accumulation, one would expect a reduction in the e.m.f. shift from peak to steady-state currents when the cell was immersed in high K^+ Ringer. This follows from the observation that current reduction due to a change in e.m.f. depends upon the percentage change in concentration near the membrane: for a given magnitude of maximum outward current (that is a particular rate of efflux of K^+), the percentage change in K^+ concentration due to efflux from the membrane for normal Ringer is much larger than for high K^+ Ringer. It is this change in external concentration near the membrane that produces the delayed current decrease. The expectation would be that in high K^+ saline the amount of delayed current decrease for any given magnitude of peak outward current should be significantly less than for the same peak outward current in normal Ringer.

To produce a meaningful experiment, the clamping potential in high K^+ must be adjusted so maximum outward current is the same in high K^+ and normal ringer. The I–V relationship produced in an experiment of this sort is shown in Fig. 5 where 100 mM-K^+ Ringer was applied to the cell. The cell potential was held at the resting potential of the cell in both solutions. In 100 mM-K^+, this holding potential was about 30 mV more positive than in normal Ringer. To avoid problems of conductance changes associated with this depolarization, the cell in 100 mM-K^+ was given a 400 msec prepulse that returned the cell to the potential in normal ringer before applying the test command. The I–V curve shows that for comparable peak outward currents, the amount of delayed current decay in high K^+ Ringer is significantly smaller than in normal Ringer. The change in reversal potential and calculated change in concentration is given in Table 3 for current records of comparable total currents (estimated by areas of comparable size on graph paper of the current–time records). The change in concentration remains nearly the same in both records, while in high K^+ Ringer the shift in EMF from the time of peak current to time of steady-state current is reduced from the same shift in normal saline.

Another manner in which it may be demonstrated that the delayed current decrease is associated with total K^+ current rather than membrane potential is by the application of tetraethyl ammonium ion (TEA). Application of TEA should block the outward K current associated with the 'delayed rectification' channel (Hagiwara & Saito, 1959). Conse-

quently, the accumulation of K^+ at the exterior surface of the membrane should be reduced and there should be little or no shift in e.m.f. associated with depolarization. Fig. 6 shows the effect of 50 mM-TEA on the $I-V$

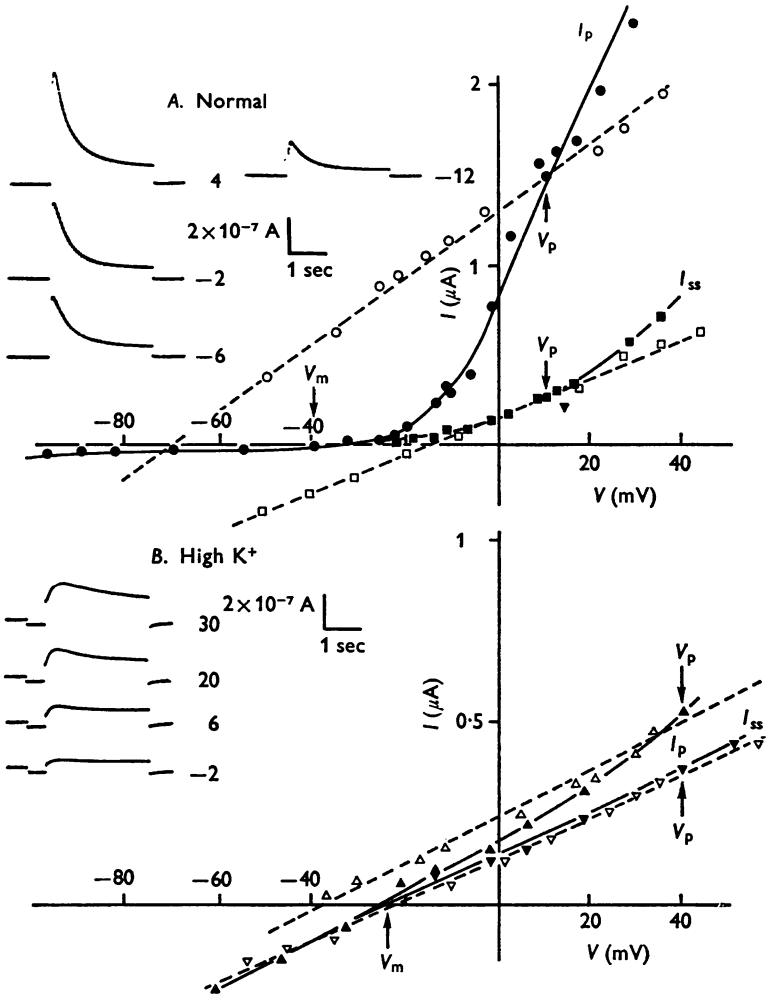


Fig. 5. Effect of high K^+ saline. The normal and instantaneous $I-V$ relationship for peak current (filled and open circles respectively) and steady-state currents (filled and open squares) are represented in A for experiment similar to that shown in Fig. 1. In B, the $I-V$ relationships of the same cell in high K^+ saline is shown. Peak normal and instantaneous currents are represented by filled and open triangles respectively, while steady-state normal and instantaneous currents are given by filled and open inverted triangles. Current–time records at various potentials are given for normal saline in the inset of A and for high K^+ saline in the inset of B. Note the difference of scale between A and B.

relationship of the cell. The reversal potential of instantaneous currents associated with the peak response has shifted in a positive direction since it no longer contains the contribution of the delayed rectifying channel

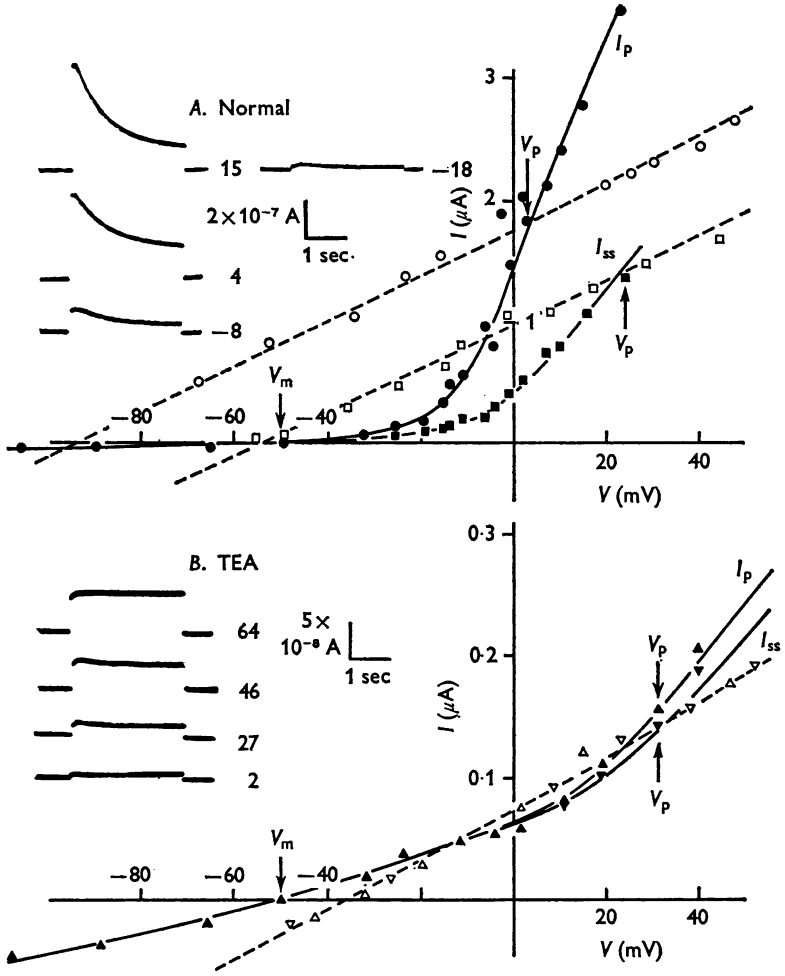


Fig. 6. Effect of TEA saline. The normal and instantaneous I–V relationship for peak current (filled and open circles respectively) and steady-state currents (filled and open squares) are represented in A for an experiment similar to that shown in Fig. 1. In B, the I–V relationships of the same cell in TEA saline is shown. Peak normal and instantaneous currents are represented by filled and open triangles respectively, while steady-state normal and instantaneous currents are given by filled and open inverted triangles. Current–time records at various potentials are given for normal saline in the inset of A and for TEA saline in the inset of B. Note the difference of scale between A and B.

(whose E_R is approximately -85 mV). But, as expected, the TEA produces a drastic reduction in the delayed current decrease and virtually eliminates the shift of reversal potential. These experiments and the low Q_{10} of the process support the hypothesis that the shift of e.m.f. is due to a diffusion limited accumulation of K^+ near the surface of the cell.

TABLE 3. Change in e.m.f. and change in K^+ concentration. In this Table values for the e.m.f. shifts for two cells in which conductance measurements were made at comparable current densities are given. In high K^+ Ringer even though the e.m.f. shift is significantly reduced the change in concentration remains approximately the same

		e.m.f. (mV) (peak)	e.m.f. (mV) (final)	Δ e.m.f. (mV)	ΔC (mM)
Saline	Normal	−72, −72	−23, −16	+49, +56	+61, +84
	High K^+	−14, +5	+1, +20	+15, +15	+82, +82

DISCUSSION

According to the Na theory (Hodgkin & Huxley, 1952*b*), when an excitable membrane is depolarized, there is a rapid increase in conductance to some ion like Na^+. This is followed by a slower increase in K^+ conductance with inactivation of the Na^+ conductance. This large increase in conductance to K^+ means that the majority of the membrane current is carried by K^+. This is the situation that is present during all but the very earliest part of the long duration voltage clamps described in these experiments. Although K^+ is the majority current carrier in the membrane, in the external solution the concentration of K^+ is relatively low so the majority of the current is carried by other ions such as Na^+ and Cl^-. Because of this difference in transport number for K^+ between the membrane and the external solution, K^+ carries current through the membrane but is then left to accumulate at the external membrane surface as most of the current is carried away by other ions.

In *Aplysia* pacemaker cells, the experiments in this paper indicate that the delayed current decrease and the associated shift of e.m.f. is due primarily to an accumulation of K^+. However, these experiments do not preclude other changes taking place in the membrane with similar time courses, but only suggest that the K^+ accumulation is the most important component of the delayed current decrease. In fact, it is fairly clear from Table 2 that there is some conductance change taking place between the time of peak and steady-state currents. This may be attributed to a small amount of K^+ inactivation such as that described in several other preparations (Nakajima & Kusano, 1966; Ehrenstein & Gilbert, 1966). However, the size of the change (about 20–30 %) is not large enough to cause a major contribution to the large change in current observed during

the delayed current decrease. In addition, there may well be a secondary K channel that is activated at long times similar to that described in the giant cell (Junge & Brodwick, 1970). But at the large command potentials and high current densities occurring in the present experiments the effects of this pathway would be relatively small.

Also it should be added that in the situation where the membrane has a moderate Cl^- conductance, some accumulation of Cl^- will take place at the internal surface of the membrane. In *Aplysia* this effect should be relatively small since the Cl^- conductance is at least an order of magnitude smaller than the K conductance at potentials where the delayed current decrease appears.

The problem remains to explain the apparent differences between the pace-making and non-pace-making cells of this preparation, that is, why the pace-makers should accumulate K^+ near their surfaces while non-pace-makers have little, if any, accumulation (Alving, 1969). The most likely hypothesis explaining this difference involves consideration of previous anatomical work done on this preparation (Bullock, 1961; Rosenbluth, 1963; and Coggeshall, 1967). There is some suggestion that cells such as R_{15} may have significantly more infoldings of their surface membranes than non-pace-making cells.

Consequently, these cells should accumulate K^+ more readily than non-pace-making cells; for example, R_2 (the giant cell) which usually does not display a delayed current decrease.

An estimate of the relative amounts of membrane surface contained in the infoldings can be made by considering the apparent capacitance per unit area of membrane for a pace-maker like R_{15} and a non-pace-maker like R_2 of comparable size. If it is assumed that the real capacitance of a small unit area of membrane is the same for all cells in *Aplysia*, then capacitances larger than expected represent an underestimation of the surface area of the cell. The value of apparent capacitance for R_{15} with a diameter of 210 μ is 47 ± 5 μF-cm^{-2} (S.D. of five measurements). Bullock & Horridge (1965) report a value for giant cell (R_2) 178 μ in diameter as 11 μF-cm^{-2}. This suggests that the real membrane area of R_{15} is about 4 times that of R_2 even though the cells are of comparable diameters. A further suggestion that this is the case may be gained by examining peak outward currents in the giant cell and R_{15}. Utilizing recent voltage clamp data for the giant cell (Geduldig & Gruener, 1970) and assuming an average size for the giant cell (520 ± 67 μ, S.D. of twenty-one cells), the calculated current per unit area of membrane assuming a spherical cell is about three times smaller at the same potential than that found in a parabolic burster only 320 μ in diameter. For cells of comparable diameter one would expect the ratio to be larger. The agreement between the ratio of surface mem-

brane calculated from capacitance measurements would then be in fairly close agreement with that derived from peak current considerations.

Although considerations of real membrane surface areas in the cells examined appear to explain almost entirely why one cell will accumulate K^+ with a consequent delayed current decrease, and another nearby cell will not, the possibility that the actual current density at the membrane in the two cell types is different cannot be entirely ruled out. If the current density in the pacemakers were higher this would further enhance the accumulation of K^+ near the membrane. Although the calculation described above suggests that the current density is approximately the same, the question is confused by the problem of how the infoldings of the cell membrane increase as the volume of the cell increases. Since R_{15} from which the outward currents were taken above was approximately 40 % smaller than the assumed size of the giant cell, the currents are only qualitatively comparable. To answer the question density, it would be necessary to choose a pace-maker and a non-pace-maker of comparable size and determine the apparent capacity of each. Then by comparing the outward current per microfarad, the relative current densities per unit area could be judged.

The physiological significance of the delayed current decrease and its relationship to pace-maker activity is not clear at the present time. However, in conclusion it is interesting to note that although a delayed current decrease is not uncommon among these membranes studied under voltage-clamp conditions, the mechanism to produce this change varies among different species. This suggests that it may be an intrinsically useful characteristic for the cells; useful enough to promote its development in evolutionarily separated animals by a variety of different means.

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REFERENCES

ADRIAN, R. H., CHANDLER, W. K. & HODGKIN, A. L. (1970*a*). Voltage clamp experiments in striated muscle fibres. *J. Physiol.* **208**, 607–644.

ADRIAN, R. H., CHANDLER, W. K. & HODGKIN, A. L. (1970*b*). Slow changes in potassium permeability in skeletal muscle. *J. Physiol.* **208**, 645–668.

ALVING, B. O. (1969). Differences between pacemaker and non-pacemaker neurons of *Aplysia* on voltage-clamping. *J. gen. Physiol.* **54**, 512–531.

BROWN, H. M., HAGIWARA, S., KOIKE, H. & MEECH, R. W. (1970). Membrane properties of a barnacle photoreceptor examined by voltage clamp technique. *J. Physiol.* **208**, 385–413.

BULLOCK, T. H. (1961). On the anatomy of the giant neurons of the visceral ganglion of *Aplysia*. In *Nervous Inhibition*, pp. 233–240, ed. FLOREY, E. New York: Pergamon.

BULLOCK, T. H. & HORRIDGE, G. A. (1965). *Structure and Function in the Nervous Systems of Invertebrates.* San Francisco, California: Freeman.

CARPENTER, D. O. & ALVING, B. O. (1968). A contribution of an electrogenic sodium pump to membrane potential in *Aplysia* neurones. *J. gen. Physiol.* **52**, 1–21.

COGGESHALL, R. E. (1967). A light and electron microscope study of the abdominal ganglion of *Aplysia californica*. *J. Neurophysiol.* **30**, 1263–1287.

CONNOR, J. A. & STEVENS, C. F. (1971). Inward and delayed currents in isolated neutral somata under voltage clamp. *J. Physiol.* **213**, 1–20.

EHRENSTEIN, G. & GILBERT, D. L. (1966). Slow changes of potassium permeability in the squid giant axon. *Biophys. J.* **6**, 553–566.

FRANKENHAEUSER, B. & HODGKIN, A. L. (1956). The after-effects of impulses in the giant nerve fibres of *Loligo*. *J. Physiol.* **131**, 341–376.

FRAZIER, W., KANDEL, E., KUPFERMANN, I., WAZIRI, R. & COGGESHALL, R. (1967). Morphological and functional properties of identified neurons in the abdominal ganglion of *Aplysia*. *J. Neurophysiol.* **30**, 1288–1351.

GEDULDIG, D. & GRUENER, R. (1970). Voltage clamp of the *Aplysia* giant neurons: early sodium and calcium currents. *J. Physiol.* **211**, 217–244.

GEREN, B. B. & SCHMITT, F. O. (1954). The structure of the Schwann cell and its relation to the axon in certain invertebrate nerve fibres. *Proc. natn. Acad. Sci. U.S.A.* **40**, 863–870.

GORMAN, A. L. F. & MARMOR, M. F. (1970*a*). Contribution of the sodium pump and ironic gradients to the membrane potential of a molluscan neurone. *J. Physiol.* **211**, 897–918.

GORMAN, A. L. F. & MARMOR, M. F. (1970*b*). Temperature dependence of a sodium-potassium permeability ration of a molluscan neurone. *J. Physiol.* **211**, 919–932.

HAGIWARA, S., KUSANO, K. & SAITO, M. (1961). Membrane changes of *Onchidium* in potassium rich media. *J. Physiol.* **155**, 470–489.

HAGIWARA, S. & SAITO, N. (1959). Voltage–current relations in nerve cell of *Onchidium verruculatum*. *J. Physiol.* **148**, 161–177.

HAGIWARA, S., TAKAHASHI, K. & JUNGE, D. (1968). Excitation-contraction coupling in a barnacle muscle fiber as examined with voltage clamp technique. *J. gen. Physiol.* **51**, 157–175.

HODGKIN, A. L. & HUXLEY, A. F. (1952*a*). Currents carried by sodium and potassium ions through the membrane of the giant axon of *Loligo*. *J. Physiol.* **116**, 449–472.

HODGKIN, A. L. & HUXLEY, A. F. (1952*b*). Components of membrane conductance in the giant axon of *Loligo*. *J. Physiol.* **116**, 473–496.

HODGKIN, A. L. & HUXLEY, A. F. (1952*c*). A quantitative description of membrane current and its application to conduction and excitation in nerve. *J. Physiol.* **117**, 500–544.

JUNGE, D. & BRODWICK, M. (1970). Post-stimulus potassium conductance increase in *Aplysia* giant neurons. *Physiologist, Wash.* **13**, 237.

KIKUCHI, R., NAITO, K. & TANAKA, I. (1962). Effect of sodium and potassium ions on the electrical activity of single cells in the lateral eye of the horseshoe crab. *J. Physiol.* **161**, 319–343.

NAKAJIMA, S. & KUSANO, K. (1966). Behaviour of delayed current under voltage clamp in the supramedullary neurons of puffer. *J. gen. Physiol.* **49**, 613–628.

NAKAJIMA, S. & TAKAHASHI, K. (1966). Post-tetanic hyperpolarization and electrogenic Na^+ pump in stretch receptor neurone of crayfish. *J. Physiol.* **187**, 105–127.

RAPOPORT, S. (1970). The sodium-potassium exchange pump: relations of metabolism to electrical properties of the cell. *Biophys. J.* **10**, 246–259.

ROSENBLUTH, J. (1963). The visceral ganglion of *Aplysia californica*. Z. *Zellforsch. mikrosk. Anat.* **60**, 213–236.

STRUMWASSER, F. (1965). The demonstration and manipulation of a circadian rhythm in a single neuron. In *Circadian Clocks*, pp. 442–462, ed. ASCHOFF, J. Amsterdam: North Holland.

TRAUTWEIN, W. & KASSEBAUM, D. G. (1961). On the mechanism of spontaneous impulse generation in the pacemaker of the heart. *J. gen. Physiol.* **45**, 317–330.